Md. Jaied Hasan

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Academic Credentials

Bangladesh University of Engineering and Technology (BUET), Dhaka — Jan 2022 – Jun 2026
B.Sc. in Electrical and Electronic Engineering — **CGPA: 3.80/4.00** (Top 27%)
Major: Communication and Signal Processing — *Last four semesters: 3.85/4.00*

Research Interests

- Wi-Fi 8 (IEEE 802.11bn) and Multi-AP Coordination
- Coordinated Beamforming and Interference Management
- Machine Learning for Wireless Networks
- Network Traffic Forecasting and Capacity Planning
- Wireless Network Simulation and Optimization
- Privacy-Preserving Distributed Learning

Research Experience

Research Assistant – Department of EEE, BUET — Oct 2026 – Present
Supervisor: Dr. Md. Forkan Uddin, Professor, EEE, BUET

- Researching Wi-Fi 8 (IEEE 802.11bn) multi-AP coordinated beamforming (Co-BF), including how inter-cluster interference (ICI) from concurrently transmitting clusters affects SINR and throughput.
- Built an explainable discrete-event Python simulator with geometry-based path-loss channels, 4-element ULA beamforming, MRT/zero-forcing/regularized-ZF precoders, CSMA/CA with physical carrier sensing and NAV, AP pairing, and per-AP throughput accounting.
- Compared CSMA/CA with MRT, two-AP Co-BF with ICI, and an ICI-free counterfactual using beam-pattern, SINR-heatmap, and ICI-penalty diagnostics, Monte Carlo runs, and PHY validation tests.

Undergraduate Thesis: Federated Sparse PCA with Differential Privacy Using Correlated Noise
Supervisor: Dr. Hafiz Imtiaz, Professor, EEE, BUET

- Extended the CAPE correlated-noise protocol to federated sparse PCA with formal (ε, δ)-differential privacy guarantees for horizontally partitioned data.
- Evaluated privacy–utility, sparsity, reconstruction, and communication-cost trade-offs against five baselines on synthetic, MNIST, and UCI Adult data; first-author manuscript under review.

Publications

- **Journal manuscript under review (first author):** "Federated Sparse PCA with Differential Privacy Using a Correlated-Noise Mechanism." *The Journal of the Franklin Institute* (Q1).
- **Conference paper accepted (2026; first author):** "An Integrated Dual-Module Cyber Physical System for Motorcycle Safety: Utilizing Pre-Ride Safety, Crash Detection and Swift Response with Mobile Application Companion." *IEEE ICACCESS 2026* (acceptance rate 23.3%). [manuscript]
- **Conference submission (2026):** "ML-Based Mobile Traffic Forecasting for Network-Capacity Planning." Submitted to *IEEE ICECE 2026*.

Relevant Coursework

- Wireless Communications (with Lab)
- Communication Systems I and II
- Digital Signal Processing I (with Lab)
- Random Signals and Processes
- Probability and Statistics
- Continuous Signals and Linear Systems
- Engineering Electromagnetics
- Optical Communications
- Radar and Satellite Communication
- Linear Algebra
- Artificial Intelligence and Machine Learning (with Lab)
- Machine Learning Specialization (Coursera)

Selected Projects

ML-Based Mobile Traffic Forecasting and Bandwidth Planning — Code

- Forecast regional mobile-internet traffic using BTCL data with emphasis on the Savar region, working with hourly link-throughput (Gbps) series for two operators (Grameenphone and Robi).
- Engineered lag, 24-h rolling-statistic, calendar, and exogenous-event features (holidays, operator offers, public events, political gatherings, power cuts, rain); analyzed trend and seasonality with STL decomposition, FFT, and ADF/KPSS tests.
- Benchmarked XGBoost, LightGBM, Random Forest, LSTM, GRU, SimpleRNN, 1D-CNN, AR, ARIMA, and Holt-Winters models on chronological splits using MAE and RMSE; paper submitted to IEEE ICECE 2026.

LifeSaver Wearable Smart Helmet (basis of the IEEE ICACCESS 2026 paper) Code

- Built a two-module ESP32/ESP8266 system linked over ESP-NOW with helmet-wear and alcohol interlocks, IMU-based fall detection, GPS/GSM SOS messaging, and a Bluetooth Android companion app.

Frequency Division Multiplexing (FDM) Hardware Demonstration

- Constructed an analog FDM demonstration from discrete components, transmitting multiple signals simultaneously over a shared medium.

Lung Disease Classification from Respiratory Audio Code

- Built a MATLAB pipeline (resampling, normalization, 10-s segment extraction, MFCC features, KNN) that classifies lung sounds as normal, asthma, or heart failure, reaching 98.66% accuracy on a held-out 20% split.

Multilingual Toxicity Classification with Knowledge Distillation

- Distilled an XLM-RoBERTa teacher into a ∼5M-parameter byte-level BPE student (∼55× smaller) for Bangla, English, and code-mixed toxicity classification (HerWILL AI for Digital Safety Datathon 2026).

HIVE-R: Bio-Inspired GPS-Free Rescue Drone Code

- Ran camera–IMU visual odometry/SLAM and a YOLOv8 Nano human detector (∼180 ms per frame) on a Raspberry Pi 4 linked to a Pixhawk flight controller over MAVLink, relaying victim locations over radio telemetry.

Technical Skills

Programming	Python, C/C++, MATLAB
Wireless / Networking	IEEE 802.11 MAC (CSMA/CA, carrier sensing, NAV), multi-AP coordination, MRT/ZF/regularized-ZF precoding
Signal Processing	DSP, MFCC, time-series analysis
ML	PyTorch, TensorFlow/Keras, scikit-learn, XGBoost
Tools	MATLAB/Simulink, NumPy, Pandas, Linux, Git/GitHub, Jupyter/Colab, LaTeX

Professional Experience

FlyRank AI Aug 2026 – Present
Machine Learning Intern *Remote*

- Developed an ML pipeline to detect and rank declining web content across 23,000+ webpages using historical traffic and trend-based features, with Precision@K evaluation, time/group-aware validation, data-leakage audits, and error analysis.

Banglalink Digital Ltd. 18 – 25 Nov 2025
Industrial Attachment

Bangladesh Mathematical Olympiad Nov 2022 – Dec 2025
Academic Team Member

Honors and Awards

- Dean's List Scholarships (Levels 1 and 3); University Merit Scholarship; Shamsul Huda Chowdhury Scholarship; Professor Dr. Ahmad Ullah and Mrs. Chemanara Ahmad Ullah Trust Fund Scholarship.
- 1st Runner-Up, Brain of Varendra – Varendra University, Rajshahi.